

# Cloud-Based Product Management System

CASE STUDY: 11-04-2026

UID: 22BAI71166

The screenshot displays the Microsoft Azure portal interface for the 'Backend\_nsg' Network security group. The left sidebar shows navigation options: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Settings, Inbound security rules (selected), Outbound security rules, Network interfaces, Subnets, and Properties. The main content area shows a list of inbound security rules with the following columns: Priority, Name, Port, Protocol, Source, Destination, and Action. The rules are as follows:

| Priority | Name                    | Port | Protocol | Source            | Destination    | Action |
|----------|-------------------------|------|----------|-------------------|----------------|--------|
| 100      | Allow_frontend_API      | 5000 | TCP      | 10.0.1.0/24       | Any            | Allow  |
| 110      | Allow_RDP_Admin         | 3389 | TCP      | 61.246.17.215     | Any            | Allow  |
| 65000    | AllowVnetInBound        | Any  | Any      | VirtualNetwork    | VirtualNetwork | Allow  |
| 65001    | AllowAzureLoadBalanc... | Any  | Any      | AzureLoadBalancer | Any            | Allow  |
| 65500    | DenyAllInBound          | Any  | Any      | Any               | Any            | Deny   |

The bottom of the screen shows a Windows taskbar with the date 11-04-2026 and time 12:32.

The screenshot displays the Microsoft Azure portal interface for the 'Frontend\_nsg' Network security group. The left sidebar shows navigation options: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Settings, Inbound security rules (selected), Outbound security rules, Network interfaces, Subnets, and Properties. The main content area shows a list of inbound security rules with the following columns: Priority, Name, Port, Protocol, Source, Destination, and Action. The rules are as follows:

| Priority | Name                    | Port | Protocol | Source            | Destination    | Action |
|----------|-------------------------|------|----------|-------------------|----------------|--------|
| 100      | HTTP                    | 80   | TCP      | Any               | Any            | Allow  |
| 110      | HTTPS                   | 443  | TCP      | Any               | Any            | Allow  |
| 120      | RDP                     | 3389 | TCP      | 45.248.159.8      | Any            | Allow  |
| 65000    | AllowVnetInBound        | Any  | Any      | VirtualNetwork    | VirtualNetwork | Allow  |
| 65001    | AllowAzureLoadBalanc... | Any  | Any      | AzureLoadBalancer | Any            | Allow  |
| 65500    | DenyAllInBound          | Any  | Any      | Any               | Any            | Deny   |

The bottom of the screen shows a Windows taskbar with the date 11-04-2026 and time 12:32.

Microsoft Azure | smartretail | Networking

SQL server

Search

Overview  
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Quick start  
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SQL databases  
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Properties

No virtual network rules found.

Firewall rules

Allow certain public internet IP addresses to access your resource. [Learn more](#)

+ Add your client IPv4 address (61.246.17.215) + Add a firewall rule

Rule name ↓ Start IPv4 address End IPv4 address

ClientIPAddress\_2026-4-10\_14... 61.246.17.215 61.246.17.215

Exceptions

☒ Allow Azure services and resources to access this server

Save Discard

Notifications

More events in the activity log → Dismiss all

Deployment succeeded

Deployment 'Microsoft.SqlDatabase.newDatabaseExis...' to resource group 'SmartRG' was successful.

Go to re... Go to resourc...

7 minutes ago

Executed delete command on 1 selected items

Succeeded: 1, Failed: 0, Canceled: 0.

10 minutes ago

Deployment succeeded

Deployment 'CreateSqlDatabase-1775810514420' to resource group 'SmartRG' was successful.

Pin to d... Go to resou...

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14:29 10-04-2026

Microsoft Azure | SmartRG | api-vm | Connect

Virtual machine

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Native RDP

Source machine

Source machine OS Windows

Source IP address Local IP | 45.248.159.8 [Connecting over a VPN?](#)

Destination VM

VM IP address Private IP | 10.0.2.5

VM port 3389

Connection prerequisites

Line of sight to private IP address VM stopped

VM access VM stopped

Validate

Connect using RDP file

Download and open file to connect Download RDP file

Username azureuser

Forgot password? Reset password

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Microsoft Azure | SmartRG | MVC-VM | Connect

Virtual machine

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Native RDP

Source machine

Source machine OS Windows

Source IP address Local IP | 45.248.159.8 [Connecting over a VPN?](#)

Destination VM

VM IP address Public IP | 20.198.8.191

VM port 3389

Connection prerequisites

VM access VM stopped

Validate

Connect using RDP file

Download and open file to connect Download RDP file

Username azureuser

Forgot password? Reset password

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